Drug induced lung diseases

By
Maha Elkholy
Professor of pulmonology



Definition

- A common clinical problem in which a patient without previous pulmonary disease develops respiratory symptoms.

- This lung disease brought on by a bad reaction to a medicine (drug).

- Drug-induced pulmonary diseases can present in a variety of syndromes , clinical presentations , and radiographic patterns .

- The most common presentations of drug-induced pulmonary disease are an abnormality on a chest radiograph or a symptom complex

Syndromes of drug induced pulmonary disease

- Asthma
- Bronchiolitis obliterans organizing pneumonia
- Hypersensitivity infiltrate
- Interstitial pneumonitis or fibrosis
- Non cardiogenic pulmonary edema
- Pleural effusions
- Pulmonary infiltrates with eosinophilia
- Pulmonary vascular disease

Drugs causing common clinical presentations of pulmonary toxicity

Asthma
Aspirin
Beta-blockers
Cocaine
Dipyridamole
Hydrocortisone
Interleukin-2
Methylphenidate
Nitrofurantoin
Protamine
Sulfasalazine
Timolol
Vinca alkaloids (with mitomycin)

Bronchiolitis obliterans organizing pneumonia
Amiodarone
Bleomycin
Cocaine
Cyclophosphamide
Methotrexate
Mitomycin-C
Penicillamine
Radiation
Sulfasalazine
Tetracycline, minocycline

Drug-induced systemic lupus erythematosus
Diltiazem
Diphenylhydantoin
Hydralazine
Hydrochlorothiazide
Isoniazid
Minocycline
Para-aminosalicylic acid
Penicillamine
Procainamide
Propylthiouracil
Streptomycin
Sulfonamides

Hypersensitivity infiltrate
Azathioprine plus 6-mercaptopurine
Busulfan
Fluoxetine
Radiation

Interstitial pneumonia or fibrosis
Amphotericin B
Bleomycin
Busulfan
Carbamazepine
Chlorambucil
Cocaine
Cyclophosphamide
Diphenylhydantoin
Flecainide
Heroin
Melphalan
Methadone
Methotrexate
Methylphenidate
Methysergide
Mineral oil
Nitrofurantoin
Nitrosoureas
Procarbazine
Silicone
Tocainide
Vinca alkaloids (with mitomycin)

Noncardiac pulmonary edema
Beta mimetics (terbutaline, ritodrine)
Chlordiazepoxide
Cocaine
Cytarabine
Ethiodized oil
Gemcitabine
Heroin
Hydrochlorothiazide
Methadone
Mitomycin-C
Phenothiazines
Protamine
Sulfasalazine
Tocolytic agents
Tricyclics
Tumor necrosis factor
Vinca alkaloids (with mitomycin)

Parenchymal hemorrhage
Anticoagulants
Azathioprine plus 6-mercaptopurine
Cocaine
Mineral oil
Nitrofurantoin
Radiation

Pleural effusion
Amiodarone
Anticoagulants
Bleomycin
Bromocriptine
Busulfan
Granulocyte-macrophage colony-stimulating factor (GM-CSF)
Interleukin-2
Methotrexate
Methysergide
Mitomycin-C
Nitrofurantoin
Para-aminosalicylic acid
Procarbazine
Radiation
Tocolytic agents

Pulmonary infiltrate with eosinophilia
Amiodarone
Amphotericin B
Bleomycin
Carbamazepine
Diphenylhydantoin
Ethambutol
Etoposide
GM-CSF
Isoniazid
Methotrexate
Minocycline
Mitomycin-C
Nitrofurantoin
Para-aminosalicylic acid
Procarbazine
Radiation
Sulfasalazine
Sulfonamides
Tetracycline
Trazodone

Pulmonary vascular disease
Appetite suppressants (dexfenfluramine, fenfluramine, phentermine)
Busulfan
Cocaine
Heroin
Methadone
Methylphenidate
Nitrosoureas
Radiation

TABLE 4

Radiographic patterns of drug-induced lung disease

Focal alveolar opacities
Amiodarone
Nitrofurantoin
Aminosalicylic acid
Penicillin
Sulfonamides
Mineral oil aspiration

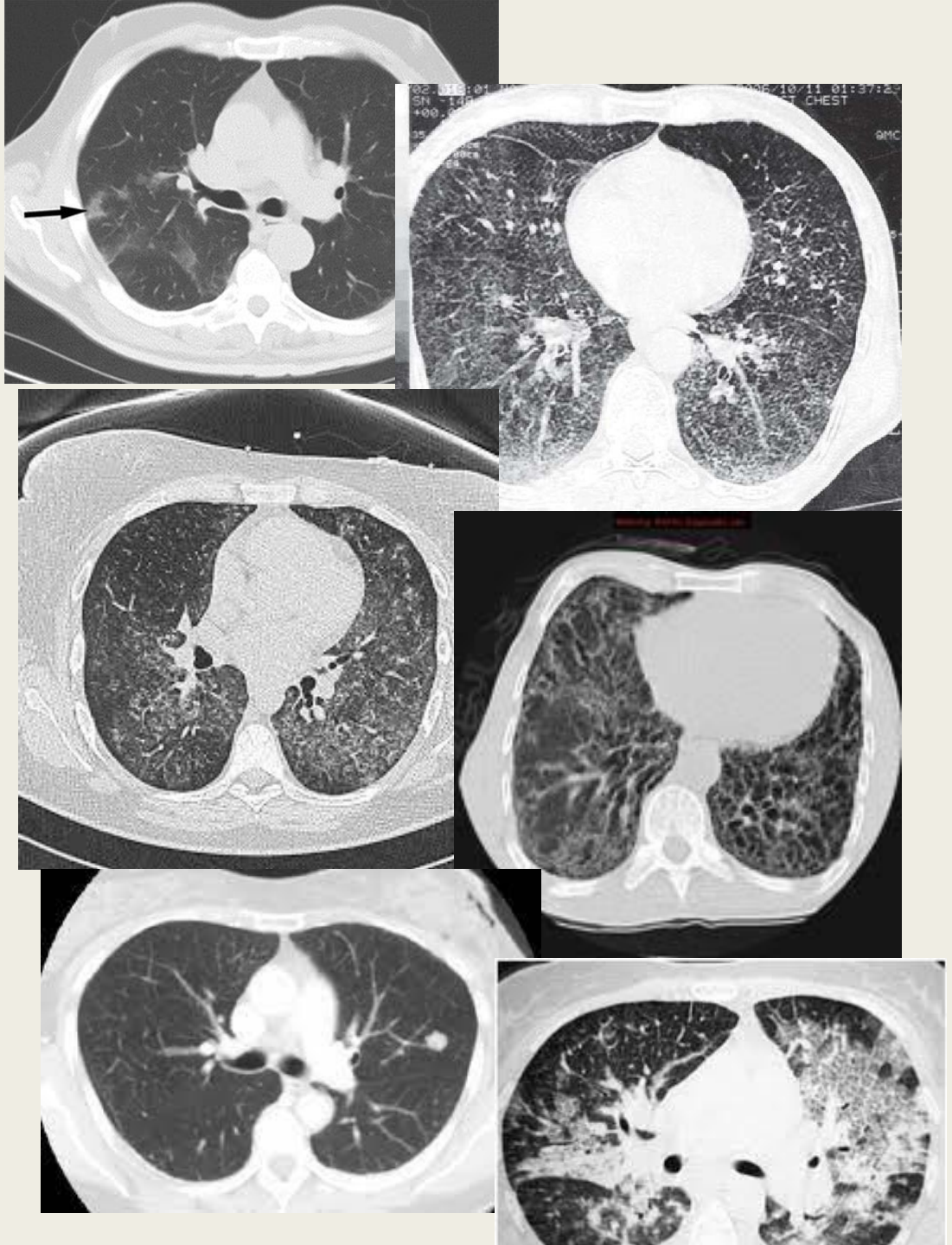
Diffuse alveolar opacities
Pulmonary edema
- Cocaine
- Cytosine arabinoside
- Heroin
- Interleukin-2
- Morphine
- OKT3
- Ritodrine
- Salicylates
- Terbutaline
- Tricyclics

Pulmonary hemorrhage
- Anticoagulants
- Cocaine
- Penicillamine
- Quinidine

Diffuse interstitial opacities
Methotrexate
Nitrofurantoin
Procarbazine
Carmustine
Bleomycin
Busulfan
Cyclophosphamide
Methotrexate
Mitomycin-C
Amiodarone
Gold salts
Oxygen
Tocainide

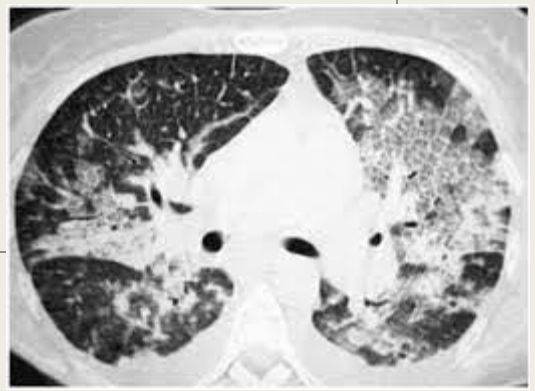
Pulmonary nodules
Amiodarone
Bleomycin
Cyclosporine
Mineral oil aspiration

Bronchospasm due to aspirin generally becomes apparent in the 3rd to 4th decade of life



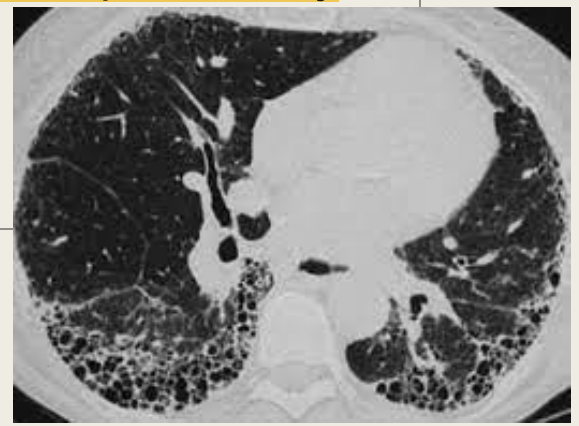
Noncardiogenic pulmonary edema

- It manifests as acute respiratory distress occurring over several hours.
- Capillary endothelial injury→ causing increased permeability edema.
- Physical examination demonstrates diffuse crackles,
- Chest roentgenograms show diffuse ill-defined acinar infiltrates and normal heart size.
- laboratory examination shows significant hypoxemia
- Cocaine and Heroin are the most common cause



Interstitial pneumonitis or fibrosis

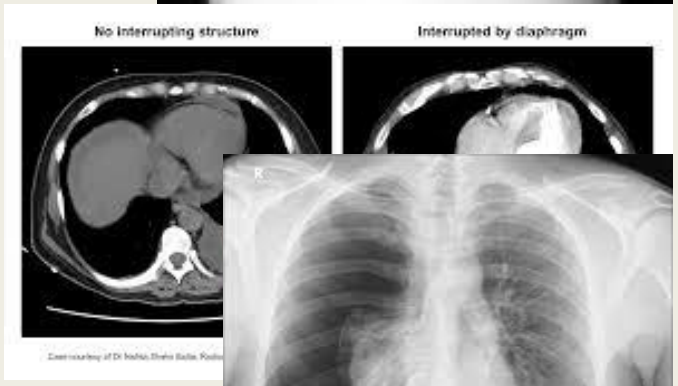
- Acute pneumonitis may present in a manner similar to noncardiogenic pulmonary edema
- In the chronic form, manifestations are a slowly progressive cough, dyspnea, weight loss, and clubbing.
- Radiographic findings are usually reticular infiltrates starting in the subpleural region of the lung bases and progressing to include the entire lung.
- The prognosis generally is worse than for drug-induced noncardiogenic pulmonary edema
- Bleomycin, cocaine , methotrexate are common causes



Pleural and mediastinal disorders



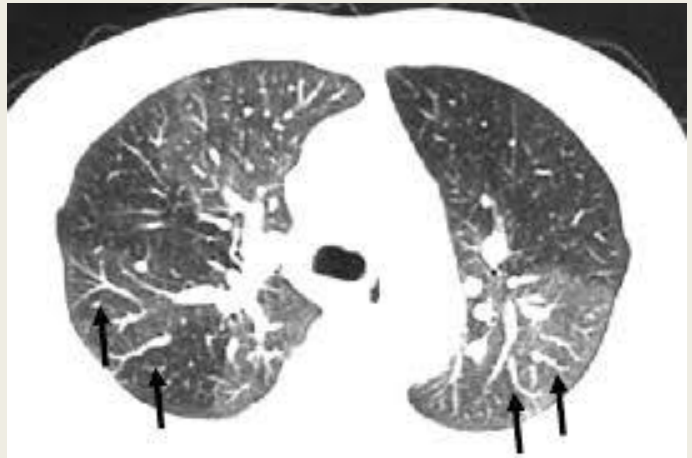
- Pleural effusions that resolve spontaneously or with discontinuation of the drug are seen in drug-induced systemic lupus erythematosus.
- Pneumothorax may complicate systemic chemotherapy and illicit drug use.
- Hemothorax or mediastinal hematoma can be a complication of anticoagulants.
- Mediastinal adenopathy may be a complication of cyclosporin, or methotrexate use.
- Mediastinal lipomatosis may complicate corticosteroid use.
- An acute pleural effusion in association with drugs can also been seen as part of a hypersensitivity reaction.







Pulmonary vascular disease



- The incidence of pulmonary veno-occlusive disease, characterized by the occlusion of pulmonary venules with subsequent elevated pulmonary arterial pressures,
- It is increased in patients receiving cytotoxic agents for the treatment of malignancy.
- The diagnosis usually requires an open lung biopsy.
- Illicit drugs can cause angiitis and hypertension.
- Alpha adrenergic nasal sprays have been associated with interstitial fibrosis and obliteration of pulmonary vessels.
- Estrogen-containing drugs as well as appetite suppressants have been associated with development of pulmonary hypertension

Drug-induced lupus

- Symptoms include fever, myalgias, rash, arthralgias, arthritis, and serositis.
- Patterns of response can be pleural effusion with or without pleuritic pain, diffuse interstitial pneumonitis, and alveolar infiltrates.
- Important features seen in drug-induced lupus can help differentiate it from spontaneous lupus. The female-to-male ratio is 1:1 in drug-induced lupus, but 9:1 in spontaneous lupus.
- Manifestations that are rare in drug induced lupus include discoid lesions (occurring in 0% of cases in one series), malar erythema (2%), renal disease (5%), and central nervous system disease (0%).
- Complement and immune complexes are normal in drug induced lupus and elevated in spontaneous lupus.
- In more than 95% of cases of drug induced lupus, antihistone antibodies are present but other autoantibodies are absent, including those directed against doublestranded DNA.
- 90% cases of drug-induced lupus are caused by diphenylhydantoin, hydralazine, isoniazid, or procainamide

Drug-induced bronchospasm

- Evidence of increased airway resistance is seen in normal persons and patients with asymptomatic asthma who use propranolol and other beta-adrenergic antagonists.
- These agents should be avoided in all patients with known obstructive lung disease whenever possible.
- Aspirin produces bronchospasm in some asthmatic patients, and similar symptoms are seen with other nonsteroidal antiinflammatory agents.
- Interestingly some inhalational preparations used in the treatment of bronchospasm (eg, albuterol) can induce cough or bronchospasm because of materials other than the bronchodilator agent in the preparation.

How aspirin and nonsteroidal anti-inflammatory drugs may worsen asthma

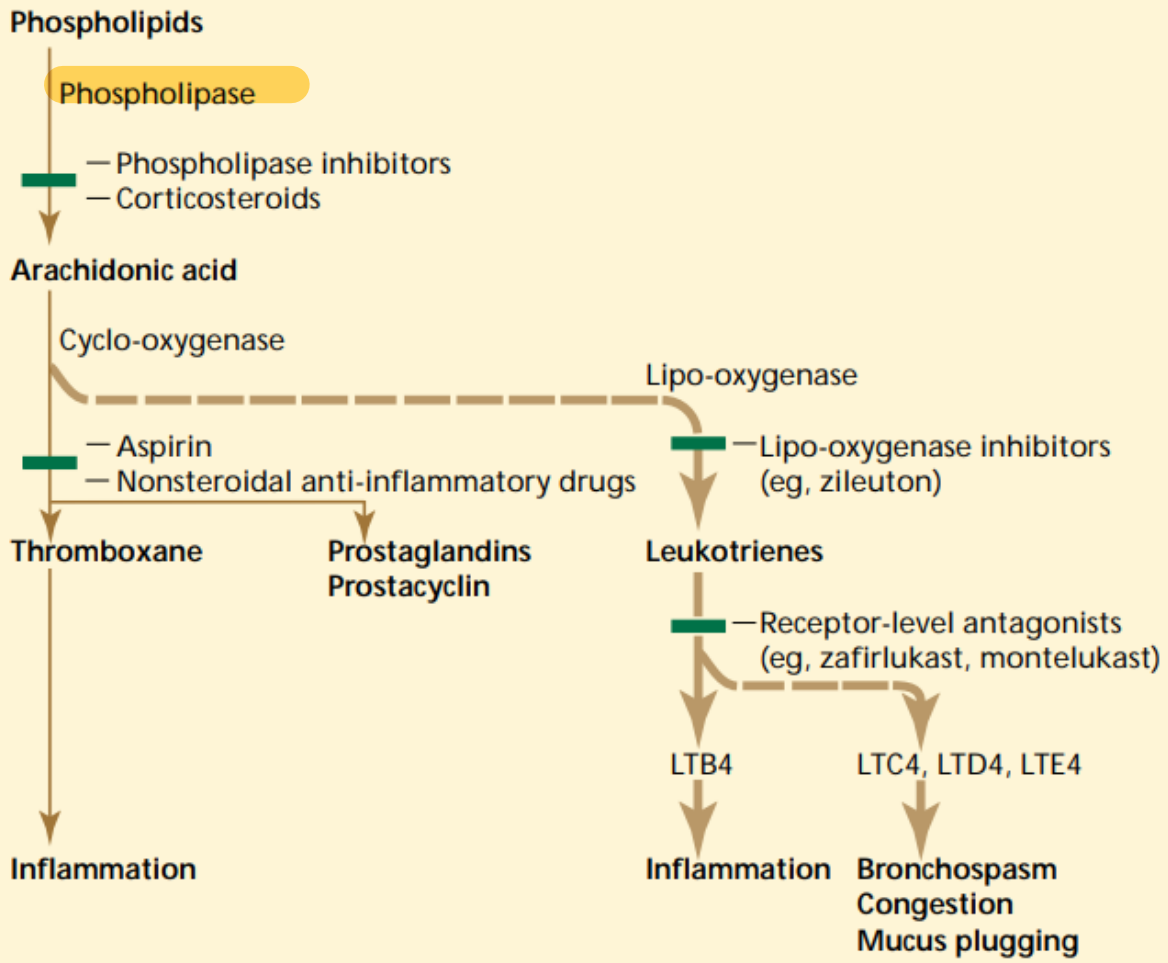


FIGURE 3. Aspirin and nonsteroidal anti-inflammatory drugs inhibit cyclo-oxygenase activity, diverting arachidonic acid to the alternate lipo-oxygenase pathway, which converts arachidonic acid to potent bronchoconstrictor mediators: leukotrienes, LTC4, LTD4, and LTE4. Lipo-oxygenase inhibitors and receptor-level antagonists may counteract the effect.

Why do angiotensin-converting enzyme (ACE) inhibitors cause cough? A possible explanation

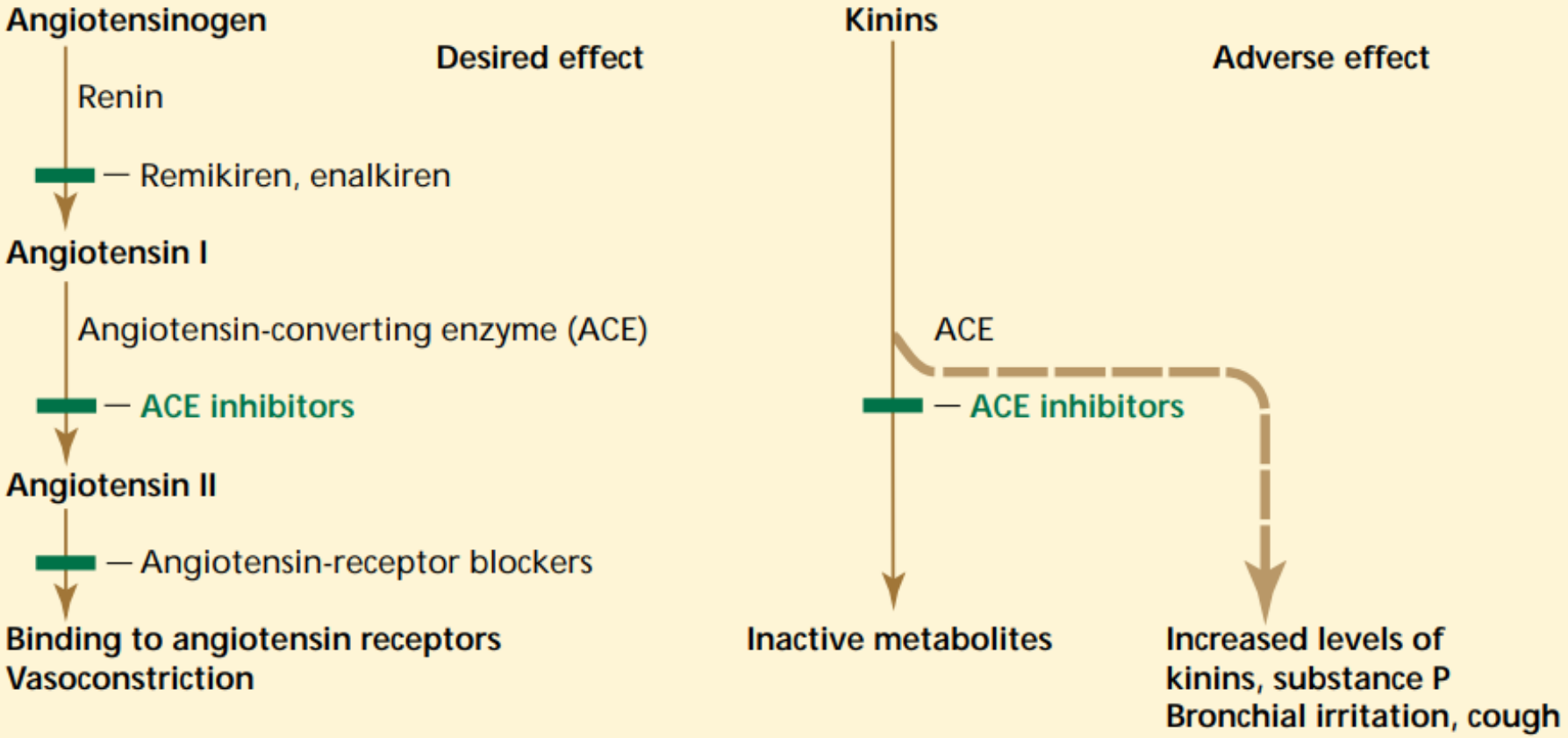


FIGURE 2. ACE inhibitors have the favorable effect of blocking conversion of angiotensin I to angiotensin II (left), but also lead to accumulation of kinins and substance P (right), which may explain the side effects of bronchial irritation and cough associated with this class of medication. Newer agents (angiotensin II receptor blockers, remikiren, enalkiren) in theory would not be associated with cough.

Treatment

- The treatment of drug-induced lung disease consists of **immediately discontinuing the offending drug and appropriately managing the pulmonary symptoms**.

- Acute episodes of drug-induced pulmonary disease usually disappear 24-48 hours after the drug has been discontinued, but chronic syndromes may take longer to resolve


Thank you